

ORIGINAL RESEARCH OPEN ACCESS

A Quantum Key Distribution System for Mobile Platforms With Highly Indistinguishable States

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Received: 28 November 2024 | **Revised:** 1 November 2025 | **Accepted:** 3 November 2025

Handling Editor: Claudio Cicconetti

Keywords: photons | quantum communication | quantum information

ABSTRACT

Quantum key distribution (QKD) allows two users to exchange a provably secure key for cryptographic applications. In prepare-and-measure QKD protocols, the states must be indistinguishable to prevent information leakage to an eavesdropper performing a side-channel attack. Here, we measure the indistinguishability of quantum states in a prepare-and-measure three-state BB84 polarisation-based decoy state protocol using resonant-cavity light-emitting diodes (RC-LEDs) as the source in the transmitter. We make the spatial, spectral and temporal DOF of the generated quantum states nearly indistinguishable using a spatial filter single-mode fibre, a narrow-band spectral filter and adjustable timing of the electrical pulses driving the RC-LEDs, respectively. The sources have fully indistinguishable transverse spatial modes. The measured fractional mutual information between an assumed eavesdropper and the legitimate receiver is 2.39×10^{-5} due to the spectral distinguishability and 4.31×10^{-5} for the temporal distinguishability. The source is integrated into a full QKD system operating in a laboratory environment, where we achieve a secure key rate of 41.5 kbits/s with an average quantum bit error rate of 2.9%. The low system size, weight and power make it suitable for mobile platforms such as uncrewed aerial vehicles (drones) or automobiles.

1 | Introduction

Ensuring secure communication is crucial in our ever-growing interconnected world, yet it remains consistently threatened by eavesdroppers who exploit system vulnerabilities to design novel attacks on our communications. The advent of quantum computers further compounds this issue because it is predicted that they can be used to compromise existing public-key cryptographic methods (see, e.g., Ref. [1]). Quantum key distribution (QKD) is one possible solution to this dilemma [2, 3]. It is based on physics principles for securely sharing a random bit string between two users, which can be used as a cryptographic key for encrypting and decrypting plain-text messages.

Currently deployed QKD systems allow point-to-point communication between a sender, commonly referred to as Alice, and a receiver, referred to as Bob. As illustrated in Figure 1, Alice's transmitter consists of a classical control system, a source of quantum light and a quantum state encoder. Bob's receiver consists of a control system, a quantum state decoder and single-photon-counting detectors. Alice encodes a random bit sequence on quantum photonic states and sends them through a free-space or fibre quantum channel to Bob, who decodes and detects them. The transmitter and receiver are connected through a classical communication channel for control-system clock synchronisation, reconciliation, error correction and privacy amplification.

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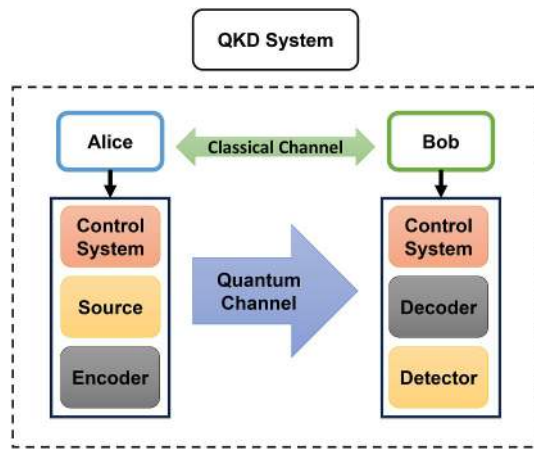


FIGURE 1 | A conceptual layout for a QKD system. Alice sends Bob quantum information through a quantum channel, and they both communicate through a classical insecure channel.

QKD systems are maturing rapidly following this general approach [4], including recent commercial systems [5–7]. These studies and systems often focus on long-distance key distribution [8–10], which will likely form the backbone of a future communication system with fixed and secure physical infrastructure. The size, weight and power (SWaP) of these systems are not a primary design metric once the transmitter and receiver can fit within a typical laboratory equipment rack.

Yet, similar to classical communication systems, different solutions are likely required for cryptographic key exchange over the ‘last mile’ between the closest backbone station and the user. Some users will be mobile so the network needs to be reconfigurable to route the quantum states between Alice and Bob. There is substantial progress on developing low-SWaP QKD systems for space [8] and flight applications [11]. Also, prototype devices aimed at hand-held applications are under development [12–15] as well as devices that can operate on small (Group 1 and 2) uncrewed aerial systems typically referred to as drones [16–18].

Developing a low-SWaP system requires a full system analysis of the trade-off between system performance and SWaP. Some recent studies have relaxed system security requirements to achieve SWaP metrics with the idea that these attack vectors will be addressed in the future [12–15].

The primary purpose of this contribution is to present a system-level analysis, design and performance characteristics of a QKD system intended for mobile applications. We address some of the security issues associated with previous low-SWaP designs by making the quantum states generated by the transmitter highly indistinguishable, thus removing several side-channel attack vectors. Our intended application is QKD between drones and/or automobiles, and the system described here is the core component of the modular quantum transmitter and receiver. Using the quantum package described here, integrated in the full modular quantum transmitter and receiver systems, we have demonstrated secure key exchange on a variety of moving mobile platforms as described in Refs. [19–21].

In the first part of this paper, we present the design considerations for the QKD system, such as choosing the QKD protocol, control system and detectors. Our goal in this section is to provide a guide for the type of questions and challenges that arise when designing such a platform. We then discuss our experimental setup to demonstrate how the selected components are assembled and managed and how data are generated, collected and processed. Finally, we present laboratory bench results of a full QKD system and discuss future improvements to our setup.

2 | Design Considerations

Here, we discuss the high-level design considerations in developing our QKD system, mentioning a few alternative solutions and discussing our final design choices. To the extent possible, we restrict our attention to commercially available components to shorten the design cycle.

2.1 | QKD Protocol

Likely the most important metric for a QKD system is the secure key rate, which requires near-ideal quantum state preparation and measurement, no leakage of information into other degrees-of-freedom (DOF) of the quantum photonic wavepackets and usually requires a high transmitter and receiver clock rate. Prepare-and-measure (P&M) [22] and entangled-state protocols [23] are the most widely used QKD schemes, where entangled-state systems are more resilient to noisy channels but typically have lower key rates than P&M QKD. We select the P&M protocol for its higher rate, which will reduce the connection time of the mobile platforms. Furthermore, we focus on free-space optical channels because they are best suited to mobile users.

The most widely used discrete-variable P&M QKD scheme is a cryptographic protocol developed by Bennett and Brassard in 1984 known as BB84 [24]. The protocol relies on transmitting quantum photonic wavepackets in which the information is encoded in one or more of the photon DOF. The most common approaches use the polarisation or time-bin/phase states, where the latter requires using delay interferometers [25]. Low-SWaP integrated photonic delay interferometers are available, but these are single-spatial-mode devices, thus substantially lowering the collection efficiency (e.g., we observe an $\sim 60\times$ reduction in received photons when we replace the multimode collection fibres with single-mode fibres) and placing much more stringent requirements on the relative alignment of the QKD transmitter and receiver. Multimode delay interferometers are being developed [26], but they are currently bulky and require high-precision optics, making them less suitable for low-SWaP systems.

Polarisation-based protocols eliminate the need for interferometers, and multimode quantum state decoders are readily available to enhance light collection efficiency. Thus, a polarisation-based protocol is more suited for applications requiring high mobility and low SWaP. However, we must select

the states to minimise errors due to the relative rotation of one platform relative to the other.

In many polarisation-based applications, a single source passes through a polarisation modulator to encode the quantum states. However, bulk modulators require high voltages and have low bandwidth for devices working in the visible and near-infrared spectrum. For this reason, we consider a transmitter with an independent source for each quantum state.

Multisource P&M protocols require indistinguishable sources to prevent side-channel attacks. In most cases [12–15], this problem is acknowledged but not addressed or quantified. In this work, we make the states as indistinguishable as possible and quantify the associated information leakage to an eavesdropper, referred to as Eve, by estimating the mutual information between Eve and Bob.

Moreover, the classic BB84 protocol relies on assuming perfect single-photon sources. Low-SWaP single-photon sources are not yet commercially available, so we focus on imperfect sources that sometimes produce multiphoton wavepackets. For these wavepackets, Eve can perform a photon number splitting (PNS) attack [27], obtaining potentially all information in these wavepackets. To mitigate this security risk, we use a decoy-state protocol developed to ensure secure communication with imperfect sources [28].

An alternative protocol is to use continuous-variable quantum key distribution, which modulates a source with Gaussian noise to encode information and does not require single-photon-counting detectors. However, it requires a single-spatial-mode receiver and mixing the received light with a locked local oscillator (LO), which requires complex pointing-and-tracking and LO stabilisation systems. See Ref. [29] for a recent free-space continuous-variable quantum key distribution system. This increased complexity does not currently match with the low-SWaP design we consider here for mobile applications, and we do not consider this protocol further.

In summary, we find that a free-space discrete-variable P&M polarisation-based BB84 scheme using multiphoton sources and the decoy state protocol is well suited for performing QKD on mobile platforms with low-SWaP restrictions.

2.2 | Control Systems

We now discuss Alice and Bob's classical control systems. There are several low-SWaP options for the control system, such as single-board computers, microcontrollers and field-programmable gate arrays (FPGAs). To narrow our choice, we need to understand a few additional requirements of a QKD system.

One requirement for the control system for a multisource QKD system operating above 10 MHz, typical for modern systems, is to generate multiple nanosecond-scale digital electrical pulses with sub-nanosecond relative timing accuracy that can be turned on in a random sequence. Single-board computers and microcontrollers usually have an operating system that does not

operate in real-time and hence has interrupts that can disrupt timing. Although some microcontrollers have independent timing/frequency units that are not disturbed by interrupts, it is difficult to have fine relative timing control across multiple channels. This suggests using an FPGA, which can precisely adjust the timing of nanosecond electrical pulses and perform tasks in parallel.

Another advantage of using an FPGA is the ability to generate true random numbers at a high rate [30]. Here, the random number generator selects in real-time the symbol and basis to encode the quantum states and the selection of decoy states. The random numbers must also be stored on the mobile platform because they form the raw key or are used in the security analysis. In Section 3.3.2 below, we discuss how FPGAs tightly integrated with a hard processor (known as a system-on-a-chip) can achieve high-throughput data storage.

We use the Intel/Terasic DE10-Standard demonstration board for Alice and Bob's control systems [31]; other system-on-a-chip devices will likely have similar performance characteristics. The FPGA part of the chip has 5.6 Mbits of embedded memory, 6 dynamically adjustable phase-locked-loop (PPL) oscillators and multiple input-output lines between the chip and standard board connectors. A high-speed bus connects the FPGA to the hard processor, a 925 MHz Dual-Core ARM Cortex-A9 hard processor using a Linux operating system with 1 GB of random-access memory and a secure digital (SD) card for long-term storage. The demonstration board has dimensions 16.7 cm × 13.0 cm × 2.7 cm, weighs 0.19 kg and, when in use during a QKD session, draws a power of 2.5 W (2.6 W) for Alice (Bob).

2.3 | Detector

For discrete-variable quantum communication systems, we must record single-photon events at Bob's receiver. We first discuss this design consideration because commercially available single-photon detectors operate over a limited spectral band and hence affect our source design discussed in the next section. Many QKD systems, including some commercial systems, use superconducting nanowire single-photon detectors [32], which have high detection efficiency and rate, and low noise. Unfortunately, they require cryogenic temperatures (~1 K), thus greatly limiting their application for mobile QKD systems, although a drone-based SNSPD system was recently reported [33].

Therefore, we focus on room-temperature single-photon avalanche photodiodes (SPADs). SPADs are available that are sensitive to photons in the telecommunication band at a wavelength near 1.55 μm , but they tend to have high noise and low detection rates unless complex custom circuits are used to improve their performance [34]. In the visible part of the spectrum, silicon-based SPADs are commercially available, have high efficiency and have a saturated detection rate greater than ~5 MHz.

Another possibility is to use a source wavelength of 1.5 μm and an up-conversion detector [35, 36]. The up-conversion detector

uses a nonlinear optical crystal pumped by a high-power laser to convert the $1.5\ \mu\text{m}$ photon to the visible or near-infrared part of the spectrum, which can be sensed by a silicon SPAD. However, the up-conversion process adds noise photons [37], only operates with a single spatial mode and would add substantially to the system SWaP.

For these reasons, we select the Excilite SPCM-AQ4C module, which features four independent detectors, matching the polarisation-based BB84 protocol that requires measuring four states. They have a peak quantum efficiency of 0.6 at a wavelength of 650 nm and are available with prealigned multimode fibre couplers. It has dimensions of $17.8\ \text{cm} \times 14.1\ \text{cm} \times 3.4\ \text{cm}$, weighs 0.68 kg and typically draws a power of 3.3 W, dominated by the thermoelectric cooler that lowers the temperature of the diode junction, thereby reducing dark counts.

2.4 | Source

We now focus on the source for our P&M polarisation-based protocol. Several groups have reported using a separate laser for each state and basis (see, e.g., Ref. [15, 38, 39]). However, the laser spectrum tends to be narrow-band; adjusting them to be identical requires injection locking or precise spectral measurement, which may have to be repeated as the temperature changes.

Another approach is to use a single laser and a combination of electro-optic polarisation and intensity modulators. However, these are often based on lithium niobate materials that display the photorefractive effect in the visible part of the spectrum and are not commercially available below $0.7\ \mu\text{m}$ where low-SWaP single-photon counting detectors are available. Also, the polarisation modulators have a typical bandwidth of only 1 MHz, which is too slow for our application. Finally, the intensity modulators only have an on-off ratio of $\sim 20\ \text{dB}$, which will increase the quantum bit error rate (QBER) of the QKD system and require a feedback-controlled bias to maintain the extinction ratio.

On the other hand, LEDs have a much broader spectrum that can then be made nearly indistinguishable using a narrow-band spectral filter. We also spatially filter the light by coupling the LED emission into single-mode optical fibres, substantially reducing the source power. We use resonant-cavity LEDs, which have a decreased emission angle and narrower spectrum to increase the spectral brightness and hence increase the single-spatial-mode power. Some years ago, RC-LEDs were available commercially at a variety of wavelengths [40], but we are aware of only one device available commercially, from Roithner LaserTechnik: the RC650-TO46FW. This RC-LED operates at a wavelength of 650 nm, which conveniently matches the peak detection efficiency of the silicon SPADs. We use these RC-LEDs in our QKD transmitter.

Wavepackets generated by a pulsed RC-LED attenuated to an average mean photon number of 0.43 required for QKD are described by thermal statistics; hence, some pulses will have more than 1 photon. The QKD system can be secured even with

these imperfect sources by using the decoy-state protocol [41–43]. Here, Alice randomly adjusts the mean photon number of the wavepackets. After a QKD session, Alice announces the mean photon number of each transmitted wavepacket. Statistical analysis of Bob's received events allows them to place a tight bound on the fraction of wavepackets having more than one photon. Here, we use a mean photon number 0.43 for the signal state, and two decoy states with mean photon numbers 0.18 and 0. The probability of transmitting each state is nonuniform, where the signal state is most probable and the other state probabilities are adjusted to optimise the secure key rate. From the analysis, Alice and Bob decide whether the measured statistics match their expected values for the mean photon number used in the experiment; a deviation indicates the presence of an eavesdropper.

Another advantage of using RC-LEDs is that they can be driven directly by the output of the FPGA, where the typical maximum current for the RC650-TO46FW is 20 mA. Furthermore, the power emitted by the RC-LED is a linear function of the injection current, which simplifies generating the decoy states as described in Section 3.1.2.

2.5 | Encoder and Decoder

To speed up system implementation, we use a small free-space optical bench consisting of a custom quarter-inch-thick aluminium plate onto which the encoder and decoder optics are placed. On Alice's side, the encoder uses half-inch optics, a compromise between system size and weight while minimising the diffraction of the light sent through the quantum channel. Assuming an Airy pattern for the light emitted from the last optic, we estimate that we can achieve up to a 90% collection efficiency at a receiver, also with half-inch optics, located a distance of $\sim 260\ \text{m}$ from the source. Longer distance QKD systems will require larger-diameter optics on the receiver or both the transmitter and receiver if diffraction is the primary loss mechanism. The encoder dimensions are $18.8\ \text{cm} \times 15.3\ \text{cm} \times 4.0\ \text{cm}$ and weighs 1.7 kg when fully loaded with the encoder optics.

To allow our system to operate beyond the 260-m transmitter-receiver separation assumed in the analysis above, Bob's decoder uses one-inch optics to enhance collecting light from our pointing, acquisition and tracking system (see Section 3.3.1 below). The decoder dimensions are $20.0\ \text{cm} \times 16.1\ \text{cm} \times 4.0\ \text{cm}$ and weighs 1.8 kg fully loaded.

3 | Implementation

3.1 | The Transmitter

Recent research has shown that sending only three quantum states in the BB84 protocol but measuring all four achieves the same secure key rate as sending four states [44, 45]. We adopt this protocol: our transmitter uses three separate RC-LEDs driven by the FPGA to produce optical pulses attenuated to the single-photon level. The RC-LEDs are coupled to single-mode fibres, which direct the light to the polarisation-based

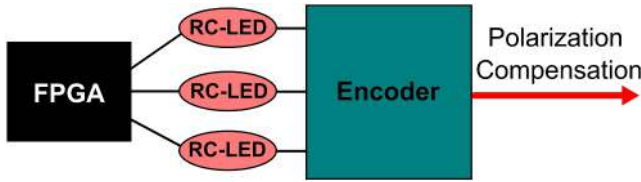


FIGURE 2 | High-level diagram showing the experimental setup for Alice's quantum transmitter.

free-space encoder (see Sections 3.1.3 and 3.1.4). After polarisation encoding, the beams are combined along a common optical axis and coupled into a single-mode fibre to make the spatial modes of each state indistinguishable. Finally, we compensate for the unitary polarisation transformation induced by the spatial-filter fibre. These subsystems are illustrated in Figure 2 and discussed in the sub-sections below.

3.1.1 | FPGA Pulse Generation

We drive the RC-LEDs using electrical pulses generated by dynamically reconfigurable PLLs on the FPGA [46]. These pulses have an amplitude of 3.3 V, a pulse width of 40 ns and a repetition rate of 12.5 MHz (the QKD system clock frequency). We use two PLLs, each with 6 outputs to drive the three RC-LEDs with the signal and nonzero decoy state. The delay of each PLL output can be adjusted with a resolution of 78 ps without recompiling the FPGA firmware. To adjust the pulse width, we combine two PLL outputs with an AND gate so that the relative delay of the resulting data pulse changes the width, where we use a typical pulse width of 10 ns. Small adjustments of the electrical pulses are made so that each state's quantum photonic wavepackets have essentially identical pulse widths as discussed in Section 3.2. We adjust the relative timing of one data pulse relative to the other by setting the delay of both PLL outputs used to generate a data pulse.

To select the transmitted state (one of the three polarisation states, signal state or decoy states), we use a digital true random number generator [30] on the FPGA that produces a 40-ns-long pulse. The pulse representing the randomly selected state and the data pulse are combined with an AND gate to unmask the desired state as illustrated in Figure 3.

The true random number generator is based on a physically unclonable function [47] and provides security beyond a pseudo-random number generator. In the future, we can upgrade to a quantum random number generator once the commercially-available low-SWaP designs have a high enough bit generation rate.

3.1.2 | FPGA-LED Interface

The electrical pulses generated by the FPGA pass to the onboard general-purpose input-output (GPIO) connector pins and then to a custom circuit board, where three RC-LEDs are attached. Each RC-LED is connected to two electrical transmission lines on the circuit board, each connected to separate GPIO pins. This

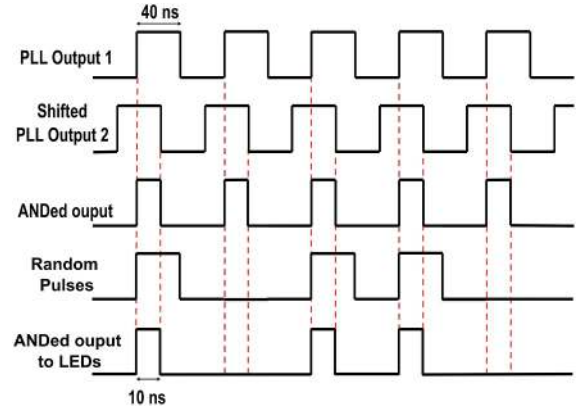


FIGURE 3 | Timing diagram illustrating the method used to generate adjustable and random electrical pulses that drive the RC-LEDs.

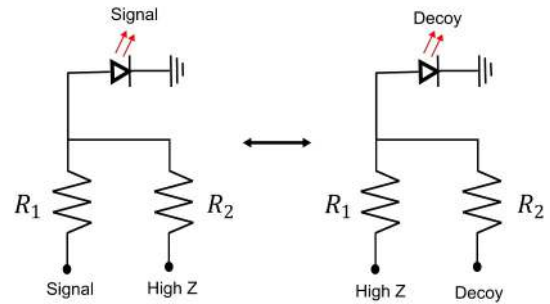


FIGURE 4 | Electrical circuit showing the different paths for the signal and decoy states.

allows us to drive a single RC-LED with two different electrical waveforms.

As shown in Figure 4 for a single RC-LED, the pulse for the signal state passes through resistor R_1 and the RC-LED to ground. The other path is set to a high impedance (high-Z) state so no current flows back to the FPGA through this path. Switching an output between the high- and low-impedance states occurs on the nanosecond timescale. To generate the decoy state, the signal path is set to the high-Z state and the decoy-state pulse passes through resistor R_2 and the RC-LED to ground. Here, $R_1 = 20.5 \Omega$ to produce an injection current of 25 mA, and $R_2 = 182 \Omega$ to produce an injection current of 10 mA so that the intensity of decoy states is $\sim 40\%$ lower than the signal states. Using this architecture for each RC-LED, we produce 3 signal states, 3 decoy states and 3 vacuum states using only 3 sources.

The probability of selecting the signal, nonzero decoy and vacuum decoy states. To this end, the true random number generator produces bits at a rate $8\times$ higher than the QKD transmitter rate so we have 8 bits available each moment a state needs to be accepted.

The 256 levels are used to select the state and the probability of generating the state. Table 1 gives the probability of generating each state. Right and left circularly polarised light represent the signal state and horizontally polarised light is for checking for the eavesdropper. We use an efficient scheme where we check for the eavesdropper with a lower percentage, which should increase the key rate.

3.1.3 | LED-Optical Setup Interface

We separate the control system and RC-LED sources from the free-space optical bench to minimise the weight of the subsystem containing the quantum state encoder. We use single-mode fibres (SMFs) to deliver the light from the source to the encoder. To improve the collection efficiency of light coupled into the fibres, we develop a coupler consisting of the RC-LED in its corresponding collimation package (RC-LED-650-02) and a fixed-focus collimation package (Thorlabs F220FC-B) to focus the light on the core of the SMF. We fit both of these collimators in a custom-built aluminium holder, which is spaced by an o-ring and attached with screws as illustrated in Figure 5. Adjusting the screws allows us to maximise the coupling efficiency into the SMF. We use an in-fibre variable attenuator (OZ Optics BB-700-11-650-4/125-S-40-3S3S-1-HYBK-0.25) to adjust the wavepacket to the single-photon level.

Figure 6 shows the experiment apparatus with the FPGA, circuit board connecting the FPGA to the RC-LEDs, couplers and SMFs. The dimension of the custom package is 6.4 cm × 5.1 cm × 2.1 cm and weighs 0.12 kg.

3.1.4 | Encoder Optical Setup

The source light is brought onto the encoder optical bench using reflective collimators (Thorlabs RC02FC-P01). To encode the states, we use a series of linear polarisers (LP), polarising beamsplitters (PBSs) and nonpolarising beam splitters (BSs) and a quarter-wave plate (QWP) to create the three quantum photonic polarisation states: $|L\rangle$, $|R\rangle$ and $|H\rangle$ as illustrated in Figure 7.

TABLE 1 | Quantum state sending fractions for the signal, decoy and vacuum states.

State	Integer fraction	Fraction (%)
Signal $ R\rangle$	121/256	47.3
Signal $ L\rangle$	121/256	47.3
Signal $ H\rangle$	2/256	0.78
Decoy $ R\rangle$	2/256	0.78
Decoy $ L\rangle$	2/256	0.78
Decoy $ H\rangle$	2/256	0.78
Vacuum	6/256	2.34

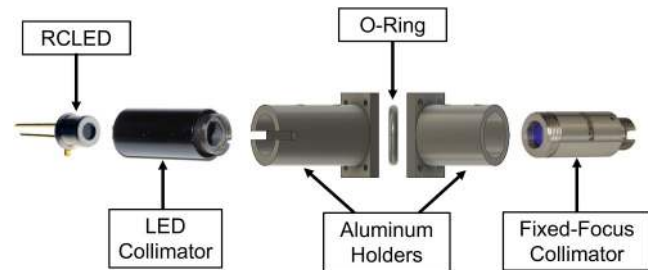


FIGURE 5 | Exploded diagram of the custom-built LED-SMF coupler.

For the circularly polarised states, the light for each channel passes through LPs, transforming the unpolarised RC-LED light into the states $|V\rangle$ (middle right of the diagram) and $|H\rangle$ (left of the diagram). These states are combined with low loss using the PBS whose output passes through a QWP with the fast axis set at 45° from the horizontal, transforming $|V\rangle$ ($|H\rangle$) to $|L\rangle$ ($|R\rangle$). The third state $|H\rangle$ is created with an LP (top right of the diagram) and combined with the $|L\rangle$ / $|R\rangle$ states using a BS with an efficiency of 50%. All states are then coupled to a SMF, subsequently passing through a series of waveplates for state compensation (see Section 3.2), and a spectral filter.

We set the mean photon number for each signal state to 0.45 at the transmitter using a SPAD detector (SAP-500, quantum efficiency 80% at 656 nm from the specification sheet) and adjusting the inline single-mode optical variable attenuators. The mean photon number for the decoy states is set relative to the signal mean photon number by the resistors discussed in Section 3.1.2 and is nominally 40% of the signal mean photon number or 0.18.

The complete transmitter setup (FPGA, RCLEDs and coupler and the encoder) measures roughly 1800 cm³, weighs 2 kg and has a power consumption of 2.5 W.

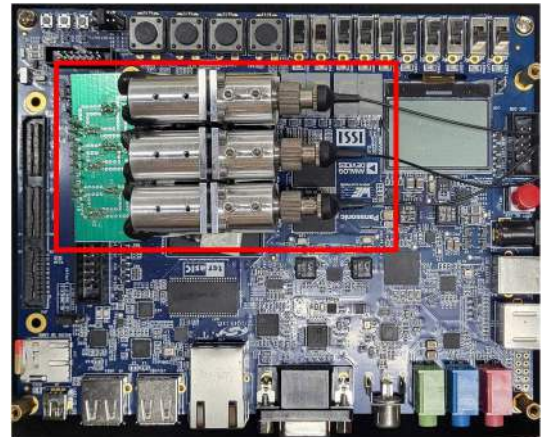


FIGURE 6 | Image showing the control system FPGA, the circuit board and the RC-LEDs with their custom-built couplers.

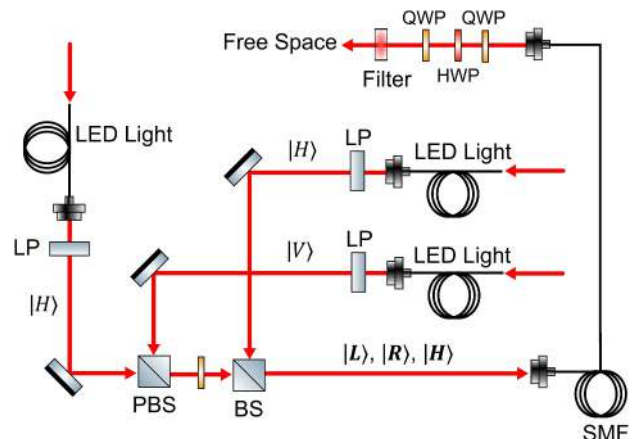


FIGURE 7 | Three-state QKD polarisation encoder showing the layout of the optical components.

3.2 | Indistinguishability

We must ensure that photons in each of the three states are indistinguishable in the degrees-of-freedom other than polarisation to protect the QKD system from side-channel attacks made directly on the transmitted quantum photonic states. For the spectral degree-of-freedom, we hand-select three RC-LEDs with the most similar spectra from a collection of 23 devices. As seen in Figure 8a, the full-width-at-half-maximum of the spectra are ~ 7 nm and the emission peaks are within ~ 2 nm. Also seen is a spectral shift for the same RC-LED when driven by a lower current to create the decoy state. Using the procedure given in Appendix A1, we find that the fraction mutual information between Eve and Bob for these states is equal to 1.77×10^{-4} with a 1.2×10^{-6} measurement bias and an uncertainty of 1.6×10^{-5} .

To increase the indistinguishability, we pass the combined states through a 1.2-nm-wide spectral filter (see Figure 7) with a centre wavelength of 656.3 nm (Andover 656FS02-12.5), producing the spectra for each state shown in Figure 8b. The fractional mutual information decreases by about an order-of-magnitude to 2.4×10^{-5} with a 2.4×10^{-5} measurement bias and uncertainty of 1.2×10^{-5} .

We next discuss the temporal indistinguishability of the quantum photonic states. Temporal variations arise from multiple

sources in our system, including differences in electrical and optical paths, and the change in the RC-LED rise time with current apparent when comparing the signal and decoy state optical waveforms. As described in Section 3.1.1, we adjust the delay of the PPL outputs to align and shape the waveforms.

We measure the optical waveform of each state by passing them, one at a time, through the entire electrical and optical sub-systems and measure the resulting wavepacket with a SPAD detector (SAP-500) with a jitter of ~ 170 ps. The resulting electrical pulse time-of-arrival T' is measured with a multi-hit time tagger (Swabian TimeTagger Ultra, with a jitter of 8 ps). We further attenuate the light to a mean photon number of ~ 0.1 to prevent waveform distortion. We also send the electrical pulse that drives the RC-LED to the time-tagger with arrival time denoted by T . The waveform is created by histogramming the number of counts for the time difference $\Delta T = T - T'$. Typically, we collect $\sim 10^8$ events for each waveform. The experimental setup used for the temporal compensation is shown in Figure 9a.

We use the following procedure to adjust the waveforms. One state, selected at random, is the reference to which all other states are compared. For the other waveforms, we compare their profiles to the reference and adjust the shift and width using the corresponding PLLs in increments of 78 ps. We typically iterate this procedure ~ 5 times.

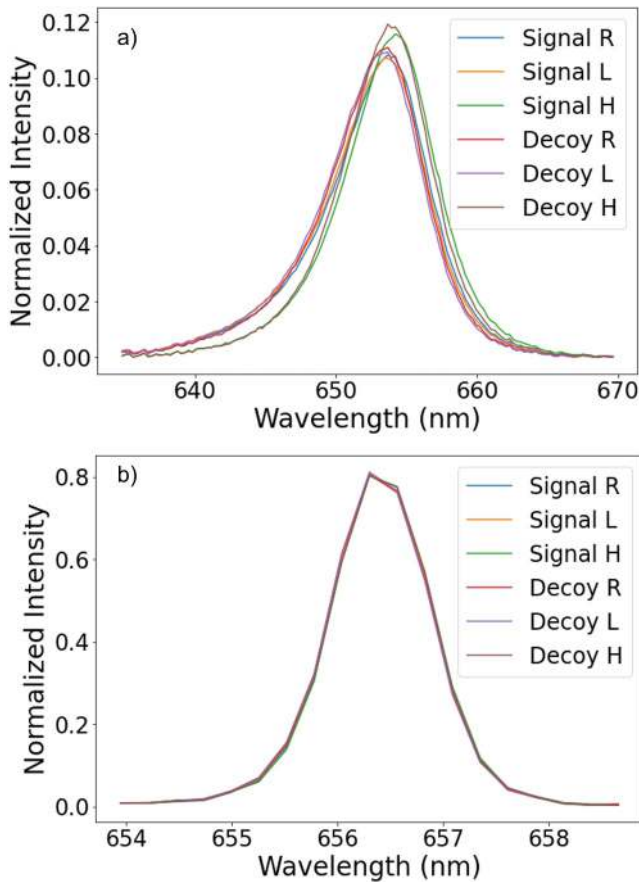


FIGURE 8 | RC-LED spectra for the different states. (a) Unfiltered and (b) filtered cases. The resolution of the spectrometer used in this measurement is 0.3 nm.

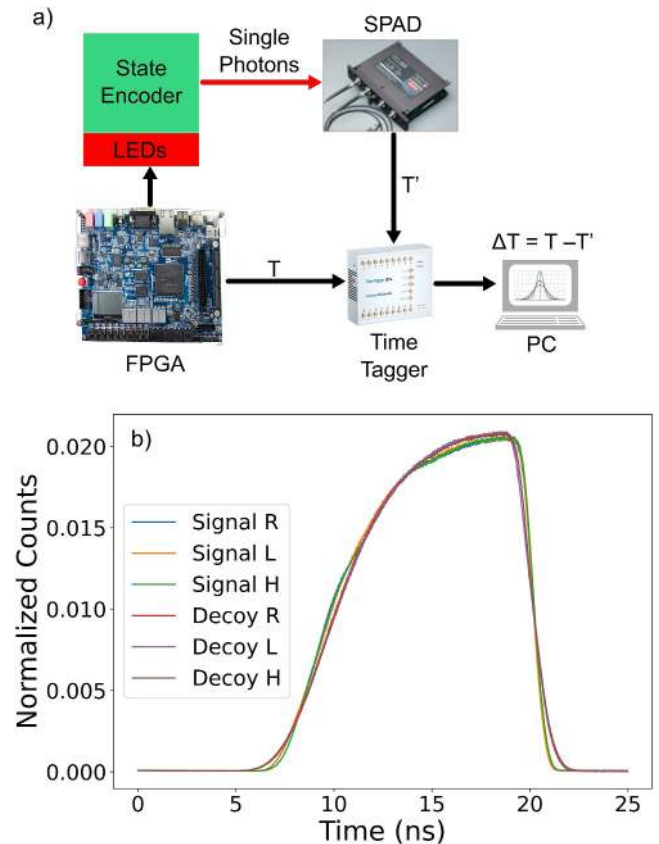


FIGURE 9 | (a) Experimental setup for measuring the temporal waveform for each of the quantum photonic states. We generate a histogram with a bin width of 20 ps. (b) Temporal profile for both signal and decoy states.

A typical final configuration is shown in Figure 9b. The decoy states have slightly different rise and fall times compared to the signal states, which is the leading contribution to the remaining distinguishability of the states. We quantify the indistinguishability of the sources using the fractional mutual information leaked to an eavesdropper that measures the spectral or temporal degree-of-freedom. The derivation in the Appendix shows how the mutual information can be calculated from the spectral or temporal probability distribution functions for each state and the sending probability of each state. For these data and modulo some assumptions of possible unresolved structure in the spectrum and waveform for the LEDs (see Appendix A), we obtain the fractional mutual information between Eve and Bob of 4.31×10^{-5} with a measurement bias of 3.18×10^{-6} and a 7.99×10^{-7} uncertainty.

Each RC-LED output beam is initially in a single transverse spatial mode because we couple it into its own SMF, but there is a possible slight misalignment of the beams on the free-space optical bench. We remove any possible distinguishability, guaranteeing that the transverse spatial modes of the states are identical, and hence indistinguishable, but coupling all three beams into a single 0.3-m-long SMF (Thorlabs 630HP, see Figure 7) before they leave the transmitter. We estimate that the critical fibre length is < 3.5 mm to have the loss of 10^6 for the least-attenuated leaky mode relative to the fundamental mode [48]. Our fibre is much greater than this length so we can coil it to remove cladding modes [12]. Finally, we use two variable focal length collimators (Thorlabs CFC11A-A) to couple in and out of the SMF. This allows us to control the outgoing beam size of the QKD signal, which is important for experiments performed at different distances.

However, passing light through an SMF, can transform the incoming polarisation states with an unknown unitary transformation, lowering the QKD system state fidelity. To reduce the change in the fibre polarisation transformation, we tape the coiled fibre to the optical bench. Even with this procedure, we find that the state of polarisation depends on temperature and the platform vibration during field experiments [19]. However, it is stable enough over the ~ 20 -min-long experiment so that we only need to compensate it at the beginning of each QKD exchange session, which we accomplish by passing the light through two quarter-wave plates and one half-wave plate (see Figure 7), adjusting them to maximise the extinction ratio of the different received states.

3.3 | QKD Receiver and Data Processing

Similar to the transmitter, the receiver has three key components illustrated in Figure 10. Received wavepackets are decoded using a free-space optical bench, each decoded state is passed to a SPAD using a multimode fibre and an FPGA is used at the control system for measuring the event time of arrival and processing the data. Below, we describe each of these subsystems.

3.3.1 | State Decoding

The optical decoder, shown in Figure 11, sorts and measures the four states of polarisation: $|L\rangle$, $|R\rangle$, $|H\rangle$ and $|V\rangle$. The incoming free-space quantum photonic states pass through a 1.2-nm-wide spectral filter (Andover 656FS02-12.5), which matches the transmitter filter and minimises noise from ambient light. Additionally, the receiver is covered by a light-proofing black plastic 3D-printed box. We use one-inch optics throughout the decoder to increase the signal collection efficiency.

The polarisations are sorted using a series of NPBS, PBS and a QWP. First, the BS randomly selects a basis to be measured (either L/R or H/V) with equal probability. On the L/R arm, we add a QWP to transform the circularly polarised light to the states $|V\rangle$ and $|H\rangle$. The PBS sorts these states and directs them to multimode fibres (Thorlabs FT200UMT, 200- μ m-diameter core, 0.39 numerical aperture), which are connected to SPAD detectors. On the H/V arm, the states are sorted by a PBS and passed to the detectors.

The receiver setup (decoder, detectors and FPGA) measures roughly 2728 cm³, weighs 2.7 kg and has a power consumption of 5.9 W.

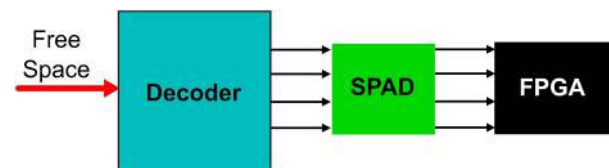


FIGURE 10 | High-level diagram showing the experimental setup for Bob's QKD receiver.

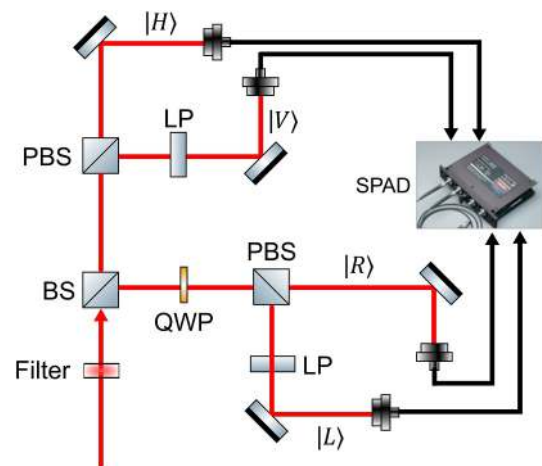


FIGURE 11 | The free-space optical setup used by Bob to decode the states generated by Alice.

3.3.2 | Data Processing

The electrical pulses generated by the SPADs are passed to the FPGA using 50-Ω-impedance coaxial cables. The voltage pulses produced by the SPADs have an amplitude of 5 V, whereas the FPGA accepts 3.3 V. We fabricated a custom resistive splitter on a small circuit board, which is inserted into the GPIO connector that reduces the voltage and impedance-matches the cable.

The received electrical pulses are time-tagged on the FPGA using a 30-bit counter incremented at every positive edge of a PLL-derived 100 MHz clock. Thus, the counter rolls over every 10.7 s. At the arrival of a detector signal, the FPGA saves the current 30-bit counter value to the FPGA's on-chip RAM.

The time-tagged data are temporarily stored in four independent 128-kB-deep banks of on-chip FPGA RAM. When any bank acquires 64 KB of data, the FPGA signals the hard processor to move that data into a 600 MB buffer located on the hard processor's 1 GB synchronous dynamic RAM (SDRAM). When the 600 MB buffer is full, the hard processor saves the data to the SD card. All events are time-tagged in parallel so that simultaneous events, such as from a multiphoton wavepacket whose photons are sorted to different detectors, are recorded. Recording simultaneous events is an important input to the system security analysis.

A similar procedure is adopted on the transmitter. We generate a new random state at a rate of 12.5 MHz. Every generated state is encoded using 3 bits and appended to a 30-bit word, which are temporarily stored in a 128-kB-deep memory bank on the FPGA. When half of the bank is full (typically every 13.1 ms), the FPGA signals the hard processor to move the 64-kB-long data segment to a 32-MB-deep buffer on its SDRAM. When the buffer is full (typically every 6.71 s), all data generation and transmission are halted for 0.5 s to allow enough time for the hard processor to save the buffer to the SD card.

Although the crystal oscillators on each FPGA demonstration board have high stability over a typical QKD session, their frequencies are slightly different and their counters do not begin simultaneously. Thus, the transmitted and received data are not temporally aligned. Synchronising Alice and Bob's data are essential for sifting, where Alice and Bob publicly discuss their basis choices and keep only the data when Bob receives a detector click and when they are prepared and measured in the same basis.

We synchronise the data using a postprocessing algorithm based on a Bayesian method [49] that recovers the clock offset between the transmitter and receiver using only the published basis and decoy state choices. Thus, it does not reveal any information to Eve because the basis and decoy state choices are already revealed over the public channel for the sifting stage of the analysis. We discard any data where the synchronisation confidence is below 95% and we only accept data within a 20-ns-wide window of the expected arrival time to reduce dark counts.

Also during postprocessing, we estimate the quantum bit-error rate (QBER) calculated for each channel ($|R\rangle$, $|L\rangle$ and $|H\rangle$) and given by

$$R_{QBER} = \frac{N_L^R}{N_L^R + N_R^R}, \quad (1)$$

$$L_{QBER} = \frac{N_R^L}{N_R^L + N_L^L}, \quad (2)$$

$$H_{QBER} = \frac{N_V^H}{N_V^H + N_H^H}, \quad (3)$$

where N_X^Y is the number of events in detector X given that Y wavepackets were sent.

4 | Tabletop QKD Results

4.1 | Raw Counts

We test our apparatus in the lab in a tabletop configuration. We first maximise RC-LED light coupled into the SMF from Section 3.1.3. We then align the optical components of Alice's side and maximise the light coupled into the spatial filter fibre from Section 3.2. We follow this by performing temporal compensation as discussed in Section 3.2. Bob's setup is aligned by back-propagating laser beams through each of Bob's multimode fibres and adjusting the optics to simultaneously overlap the beams close to the bench and at a distance of ~ 10 m.

Once both Alice and Bob's setups are aligned, we separate the transmitter and receiver by a distance of ~ 1.5 m on an optical table and align them to each other using auxiliary mirrors. We then compensate the polarisation states emitted by the transmitter as discussed in Section 3.2.

Finally, we collect data for about 100 s, which typically generates 5×10^7 events on Bob's side. Figure 12 shows the typical count rate for each state during a QKD session. Based on these data, the mean photon numbers and state sending fractions given in Section 3, we estimate the receiver loss is 15 dB, largely

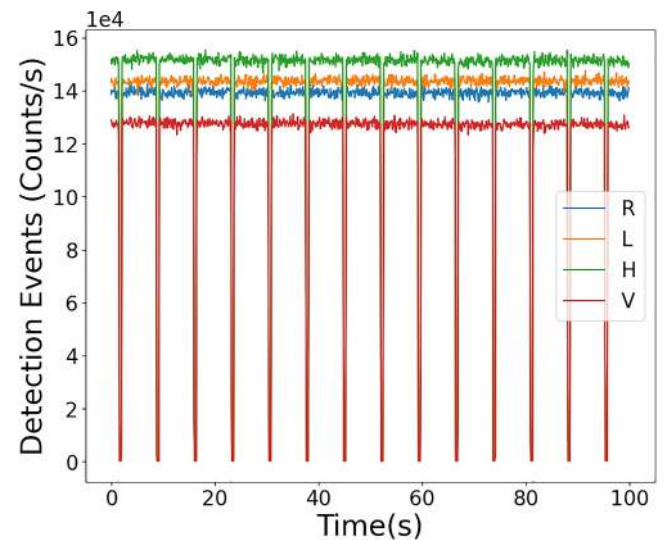


FIGURE 12 | Typical count rates on Bob's detectors for a tabletop QKD experiment.

dominated by the narrow-band spectral filter on the receiver input. Notably, the periodic drops in counts are intentional and occur when the transmitter FPGA halts all operations for 0.5 s to save data to the SD card as discussed in Section 3.3.2.

We now discuss the expected count rates. The signal states $|R\rangle$ and $|L\rangle$ have a sending fraction of 47.3% each (recall Table 1) and dominate the received counts. Because we have an approximate 50:50 passive basis choice in the receiver, we expect to observe equal counts on all detectors only considering these two states. The counts are imbalanced because of the checking state $|H\rangle$, which we send with a total fraction (signal + decoy) of 1.56%. When the state $|H\rangle$ is measured in the $|R\rangle/|L\rangle$ basis, both detectors should register the same number of counts. When it is measured in the $|H\rangle/|V\rangle$ basis, we expect more counts in the $|H\rangle$ detector and no additional counts in the $|V\rangle$ detector. This results in expected counts in the $|H\rangle$ to be the highest, the $|R\rangle$ and $|L\rangle$ detector to be lower and the same and the counts in the $|V\rangle$ detector to be the lowest. The observed count rates are within a few percent of expectations; the deviation is largely due to a small imbalance in the detected state probability when a state prepared on one basis is measured in the other.

We obtain QBERs of 3.1% in the $|R\rangle/|L\rangle$ basis and 2.7% in the $|H\rangle/|V\rangle$ basis. Here, we estimate the errors using the entire dataset shown in Figure 12, but the results will be similar using a shorter dataset given the high stability of the count rates in this tabletop setup.

4.2 | Secure Key Rate

Proving security for our QKD system is challenging [50] because we use two decoy state levels (vacuum and one intermediate mean photon number state) and send only three polarisation states to simplify the hardware design. Using one or two decoy state levels instead of three is known to produce secure key rates in the finite-length key limit is like protocols using three decoy states [51, 52]. Also, using three instead of four transmitted states [45, 53] is expected to produce the same key rate in the infinite-length key limit.

To the best of our knowledge, only Ref. [44] treats the case of two decoy states and sending three quantum states but in the infinite-length key limit. Given that our exchange session is somewhat short where finite-length key corrections may be important, we use the custom finite key analysis described in Ref. [19] that is based on a self-consistent QKD channel and a nonideal system model. Briefly, the analysis generalises the work of Ref. [53] for imperfect state generation, imperfect receiver optics and different loss and detector efficiencies along each receiver path. It combines the nonideal behaviours into a single unitary operator.

An optimiser is used to find the best fit to the model, where the inputs are obtained from quantum state tomography of the transmitted states, mean photon numbers of the signal states and a table of the QKD counts. It uses a sequential approach that first estimates part of the unitary transformation based on the signal states alone, and a second stage using all counts for

estimating the decoy state mean photon number, stray light and dark counts.

The model is incorporated into a finite-size analysis based on a composable ϵ -security framework, and errors are corrected using a symmetric blind informational reconciliation protocol [19]. The composite tolerated error is set to 10^{-8} , and the error correction efficiency factor is 1.16, where 1 represents the Shannon limit.

For the data shown Figure 12, which have $\sim 1.1 \times 10^9$ signals, the analysis finds a secure key rate of 41.5 kbp, and a total secure key of 3.9 Mb (These data are referred to as 'Run 4' in Ref. [19]). The code used in this analysis was developed within the framework of the Open QKD Security platform [54].

5 | Conclusions

We describe the design considerations for a QKD system suitable for mobile applications and use it for a laboratory QKD session. This system has been used for drone-to-drone, drone-to-car and car-to-car QKD as reported elsewhere [20, 21].

In the future, we plan several improvements to our system. Regarding SWaP, one improvement is to develop a compact, potentially monolithic encoder and decoder [14] that will reduce the SWaP and improve the alignment stability. Similarly, we found the RC-LED-to-SMF coupler misaligned over days due to the relaxation of the o-ring in the coupler. A new coupler design using prealigned optics may be beneficial.

Another bottleneck in data processing is off-line clock synchronisation. We are currently developing a method to synchronise the clocks on each FPGA to Uniform Coordinated Time using a global positioning system (GPS) timing receiver.

Author Contributions

Daniel Sanchez Rosales: data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Roderick D. Cochran:** data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Samantha D. Isaac:** investigation, methodology, resources, visualization, writing – review and editing. **Paul G. Kwiat:** conceptualization, funding acquisition, investigation, methodology, project administration, resources, supervision, writing – review and editing. **Daniel J. Gauthier:** conceptualization, data curation, funding acquisition, investigation, methodology, project administration, resources, supervision, writing – review and editing.

Acknowledgements

We gratefully acknowledge Lars Kamin and Norbert Lükenhaus for developing the custom security analysis in the context of Ref. [19], which is also applied to the dataset presented here.

Funding

We gratefully acknowledge the financial support of NASA (#80NSS-C20K0629) and the U.S. Office of Naval Research Multidisciplinary University Research Initiative program on 'Wavelength-Agile Quantum Key

Distribution in a Marine Environment' (#N00014-13-1-0627). D.S.R. and D.J.G. gratefully acknowledge the financial support of the U.S. Air Force Research Laboratory project 'Centre for Enabling Cyber Defense in Analogue and Mixed Signal Domain' (#FA8650-19-1-1741). R.D.C. and D.J.G. gratefully acknowledge the financial support of the U.S. Air Force Research Laboratory and the Southwestern Council for Higher Education (#FA8650-19-2-9300). S.D.I. and P.G.K. gratefully acknowledge the financial support of the U.S. Army Research Laboratory project 'Mobile-Platform Quantum Communication' (#W911NF2420097), the KRISS Korea project 'Development of Core Technologies for Entanglement-Based Multi-Party Hybrid Quantum Networks' (NRF #112168) and the U.S. Department of Energy project 'Advanced Quantum Networks for Science Discovery' (DOE Fermi 712869). The U.S. and Korean governments are authorised to reproduce and distribute reprints for governmental purposes notwithstanding any copyright notation thereon. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the funding agencies listed above or the U.S. or Korean governments.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All code for this project is available in ref. [55]. The Verilog code developed for this project targets the DE10-Standard FPGA board using the Quartus Prime Lite 18.1 environment. Additionally, the code to analyse the spectra, temporal profiles, tabletop and indistinguishability data are developed in Python.

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Appendix A

A1. Quantifying Indistinguishability

In our system, we use pulses generated by RC-LEDs. These pulses are not transform-limited (i.e., their spectral bandwidth is $\sim 1,000\times$ broader than the inverse pulse width). Hence, there is, in principle, undetected distinguishing information in the wavepackets. For example, there could be a unique temporal pattern in the pulses, that is, faster than the resolution of the SPADs, a fine spectral pattern below the spectrometer resolution or a frequency or phase modulation, unique to each RC-LED. An eavesdropper with a faster response-time detector, a higher resolution spectrometer or a more general pulse analysis method could use these patterns to distinguish between the three LEDs.

Here, we assume that thermal states describe the RC-LEDs without such identifying fine structure in the spectral-temporal modes, and hence the as-measured states can be used to quantify the indistinguishability. This assumption is physically reasonable because no obvious mechanisms would lead to such a fine structure. For example, although a frequency chirp in the LED spectrum could give such structure, it would need to be a different chirp for each LED to create unwanted distinguishing information, whereas the LEDs are all of the same construction and the photons pass through similar optics before leaving the transmitter.

Using our as-measured spectral-temporal measurements, we estimate the indistinguishability of Alice’s states using the mutual information to calculate the fraction of the final sifted key potentially shared between Bob and Eve due to the spectral and temporal differences. We denote E

as the set of Eve's side channel measurements, B as Bob's sifted bits and S as the statement that the channel access attempt resulted in a sifted bit. The mutual information between Eve's measurements and Bob's sifted key is given by

$$I(B; E|S) = H(B|S) + H(E|S) - H(B, E|S), \quad (\text{A1})$$

where H is the Shannon entropy

$$H(X) := -\sum_{x \in X} p(x) \log p(x). \quad (\text{A2})$$

Thus,

$$\begin{aligned} I(B; E|S) &= -\sum_k p(B_k|S) \log p(B_k|S) \\ &\quad - \sum_j p(E_j|S) \log p(E_j|S) \\ &\quad + \sum_{k,j} p(B_k, E_j|S) \log p(B_k, E_j|S). \end{aligned} \quad (\text{A3})$$

To compute the entropies, we must find $p(B_k|S)$, $p(E_j|S)$ and $p(B_k, E_j|S)$ in terms of known probabilities.

The probability $p(E_j|A_i)$ is the conditional probability of the eavesdropper measuring a state j (such as a particular wavelength in the spectral measurement or a particular arrival time in the temporal measurement), is given by the distributions in Figures 8b and 9b. The probability $p(A_i)$ is the sending probability for each state, $p(S|A_i)$, which is the likelihood of a particular state resulting in a sifted bit and $p(B_k|A_i, S)$, which is the probability of the sifted bit result recorded by Bob for a given input state. These latter three can be calculated from the sending state fraction, the expected receiver sorting and the estimated channel loss.

The conditional probability $p(B_k|S)$ can be expanded using marginalisation (i.e., summing over some variables), giving

$$p(B_k|S) = -\sum_i p(B_k|A_i, S) p(A_i|S), \quad (\text{A4})$$

where A is the set of states Alice can send. Applying Bayes' Theorem, we find

$$p(B_k|S) = -\sum_i p(B_k|A_i, S) \frac{p(S|A_i)p(A_i)}{p(S)}. \quad (\text{A5})$$

Applying marginalisation again, we find

$$p(B_k|S) = -\frac{\sum_i p(B_k|A_i, S) p(S|A_i)p(A_i)}{\sum_{\ell} p(S|A_{\ell})p(A_{\ell})}, \quad (\text{A6})$$

with all terms now being known probabilities.

The conditional probability $p(E|S)$ can be found using a similar procedure, resulting in the relation

$$p(E_j|S) = -\frac{\sum_i p(E_j|A_i, S) p(S|A_i)p(A_i)}{\sum_{\ell} p(S|A_{\ell})p(A_{\ell})}, \quad (\text{A7})$$

which can be simplified using the knowledge that $p(E_j|A_i, S) = p(E_j|A_i)$ because the distribution of Eve's measurements is fully determined by knowledge of Alice's state, and hence S is irrelevant. Using these relations, we find

$$p(E_j|S) = -\frac{\sum_i p(E_j|A_i) p(S|A_i)p(A_i)}{\sum_{\ell} p(S|A_{\ell})p(A_{\ell})}. \quad (\text{A8})$$

We can rewrite.

$$p(B_k, E_j|S) = p(B_k|E_j, S) p(E_j|S), \quad (\text{A9})$$

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, E_j, S) p(A_i|E_j, S) p(E_j|S), \quad (\text{A10})$$

$$p(B_k|A_i, E_j, S) = p(B_k|A_i, S), \quad (\text{A11})$$

because Bob's measurement outcome is fully determined by Alice's input state and the knowledge that it results in a sifted bit. Because Eve's measurement is a side channel measurement in this derivation, it has no impact on Bob's measurement in the chosen degree of freedom.

Finally, we obtain.

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, S) p(A_i|E_j, S) p(E_j|S), \quad (\text{A12})$$

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, S) \frac{p(E_j|A_i, S) p(A_i|S)}{p(E_j|S)} p(E_j|S), \quad (\text{A13})$$

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, S) p(E_j|A_i, S) \frac{p(S|A_i)p(A_i)}{p(S)}, \quad (\text{A14})$$

$$p(B_k, E_j|S) = \frac{\sum_i p(B_k|A_i, S) p(E_j|A_i, S) p(S|A_i)p(A_i)}{\sum_{\ell} p(S|A_{\ell})p(A_{\ell})}. \quad (\text{A15})$$

We have now written all relevant quantities in terms of known probabilities that can be inserted into Equation (A4) to find the mutual information Eve gains from the side-channel measurements.

However, we must also account for the systematic bias in our mutual information calculation. Because measurement noise inherently makes a variable more random, it creates a systematic bias for higher entropies. Roulston [56] shows that the observed entropy H_{obs} can be written as follows:

$$\langle H_{obs} \rangle = H_{\infty} + \sum_i \frac{\langle \epsilon_i^2 \rangle p_i}{2}, \quad (\text{A16})$$

where H_{∞} is the 'true' entropy of the system, B is the number of states and ϵ_i is the fractional measurement error in the state probability p_i , with the measured probability q_i such that

$$\epsilon_i = \frac{q_i - p_i}{p_i}. \quad (\text{A17})$$

Thus, Equation (A16) can be rewritten as follows:

$$\langle H_{obs} \rangle = H_{\infty} + \sum_i \frac{\langle (q_i - p_i)^2 \rangle}{2p_i}, \quad (\text{A18})$$

or

$$\langle H_{obs} \rangle = H_{\infty} + \sum_i \frac{\sigma_{q_i}^2}{2q_i}, \quad (\text{A19})$$

using the definition of the variance and approximating the denominator as $2q_i$ and σ_i is the error in the measurement of p_i . We now have the entropy bias in terms of known quantities, and we can add the biases of the constituent entropies to find the bias for the mutual information as a whole.

To quantify the uncertainty of the mutual information, we used two approaches. The first is the standard propagation of errors technique using derivatives given by

$$\sigma_f = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2}.$$

that is, for function f , the uncertainty σ_f in terms of the uncertainties of the constituent variable x_i and its derivatives.

However, this formula is an approximation that assumes $\partial f / \partial x_i$ is approximately constant in the neighbourhood of σ_{x_i} . Although this approximation is accurate for evaluating the temporal degree-of-freedom (for which our measurement has a smaller error), it is inaccurate for the spectral degree-of-freedom.

For the spectral data, we use a Monte Carlo sampling approach. By randomly sampling the constituent variables from their distributions and evaluating the mutual information function many times, we create a histogram representing the probability distribution of the mutual information. The standard deviation of this distribution is the uncertainty of the mutual information measurement.